

Undergraduate Recommendations

Daniele B. Provenzano | daniele.provenzano@sns.it

March 2025

Here you can find my recommendations for undergraduate-level textbooks on Classical Physics. Be sure to get the latest edition and check the errata. *Disclaimer:* these are my personal suggestions. I do not claim they are the absolute best, but they were the most suitable for me at the time of writing. This list may also change in the future.

1. The foundation of Physics is Mathematics. One of my professors used to say: “A physicist who doesn’t master mathematics is a poor physicist.” For **Mathematical Analysis** and **Calculus**, I recommend:
 - *Principles of Mathematical Analysis* (3rd ed.) by Walter Rudin.
 - *Calculus* (4th ed.) by Michael D. Spivak.
2. **Classical Mechanics** is the branch of Physics most directly connected to everyday life and, as a result, is usually the first subject introduced in school and university. The books I highly recommend are:
 - *Classical Mechanics* by John R. Taylor, for the theory.
 - *Introduction to Classical Mechanics* by David J. Morin, for the exercises. This book contains the best collection of problems in Classical Mechanics I know of. No further explanation is necessary: it is a must-have for every physicist! If you find it too challenging, you can start with its easier version: *Problems and Solutions in Introductory Mechanics*, by the same author.
 - *Physics for Mathematicians – Mechanics I*, by Michael D. Spivak. If you are looking for a book that offers a formal treatment of Classical Mechanics while remaining accessible to physicists, this is the perfect choice. It includes many insightful interludes that deepen one’s understanding of the subject, and it ranks among my top three textbooks of all time.
3. For **Waves and Oscillations**, the best resource is the draft *Waves* by David J. Morin. Although it currently lacks exercises, it provides a remarkably clear presentation of the subject. Some related problems can be found in the [book mentioned above](#).
4. Let’s move on to **Electromagnetism**. Here, the choices are straightforward:
 - *Introduction to Electrodynamics* (5th ed.) by David J. Griffiths.
 - *Electricity and Magnetism* (3rd ed.) by E. M. Purcell and D. J. Morin.

I suggest using both textbooks together. They provide plenty of exercises, but if that is not enough, I strongly recommend:

- *Problems in Classical Electromagnetism*, by A. Macchi, G. Moruzzi, and F. Pegoraro. This book offers one of the best collections of EM problems I have seen, including some particularly original ones dealing with electromagnetism in matter. I had the pleasure of attending two courses taught by Professor Macchi, and I can confirm he is an excellent teacher!

5. For **Special Relativity**, I strongly recommend the following introductory textbooks:
- *Relativity Made Relatively Easy* by Andrew M. Steane, for the theory.
 - *Special Relativity* by David J. Morin, for the exercises. This is an expanded and refined version of the chapters he wrote in [ICM](#).
6. Although Electromagnetism boasts outstanding textbooks, the same cannot quite be said for **Optics**, its natural extension. Still, I recommend:
- *Optics f2f: From Fourier to Fresnel* by C. S. Adams and I. G. Hughes.
 - *Principles of Optics* (7th ed.) by M. Born and E. Wolf.
7. Regarding **Thermodynamics**, unfortunately, I don't think there is an equivalent of Morin's or Griffiths's books for this subject. The available options are often either too theoretical or lack sufficient exercises. However, I recommend:
- *Concepts in Thermal Physics* (2nd ed.) by S. J. Blundell and K. M. Blundell.
 - *Heat and Thermodynamics* (7th ed.) by M. W. Zemansky and R. H. Dittman.
8. Among all the branches of Classical Physics, I believe **Fluid Dynamics** is conceptually the most difficult to fully grasp (and the one with the most open problems ☺). At the introductory level, I strongly recommend:
- *An Introduction to Fluid Dynamics*, by George K. Batchelor.
 - *Fluid Dynamics for Physicists*, by Thomas E. Faber.
 - *Theoretical Fluid Mechanics*, by Richard Fitzpatrick.

However, these books do not provide many introductory-level exercises.

If you are looking for a textbook that covers (almost) everything at a more advanced level, I recommend *Modern Classical Physics* by Kip S. Thorne and Roger D. Blandford.

My final suggestions are two mathematics books by the same author: *Visual Complex Analysis* and *Visual Differential Geometry* by Tristan Needham. You should definitely own at least the first one—it makes complex analysis beautifully intuitive through visual reasoning and examples.

As you can see, I have cited five works by David J. Morin in just two pages. This could mean one of two things: either I'm obsessed with him, or he's a master at writing physics textbooks. Read his books and decide for yourself.